Maths

WALT:

6Sc.03 – Understand that the mode, median, mean and range are ways to describe and summarise data sets. Find and interpret the mode, median, mean and range and consider their appropriateness in context.

WALT: Understand that the mode, median, mean and range are ways to describe and summarise data sets.



With support, I can answer questions about averages.



Independently, I can answer questions about averages.



I can complete the think like a mathematician exercise.

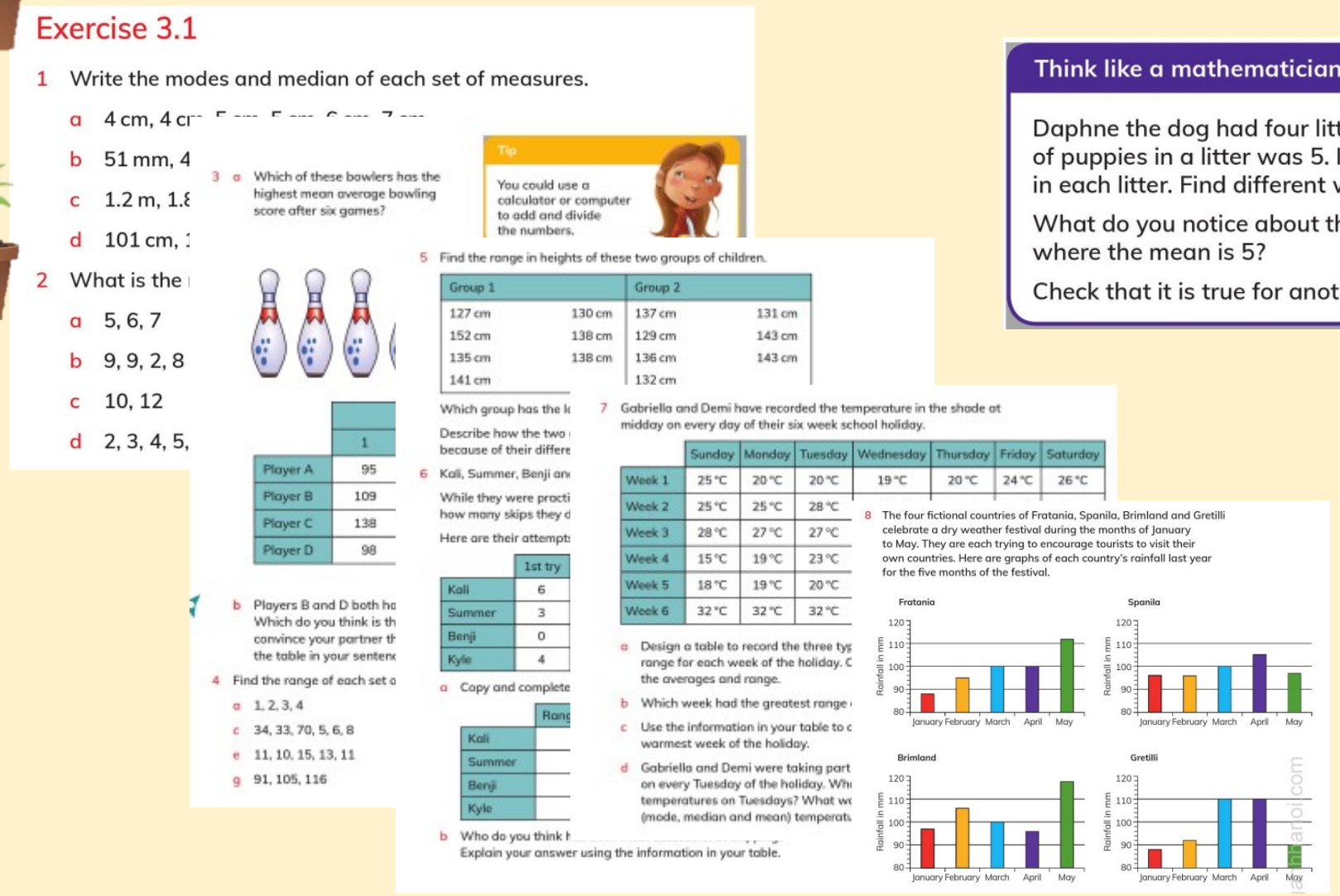
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Retrieval Practice

Write a single digit in each box to complete the statement.

6 tens + 308 hundredths + 47 thousandths = 6 3 . 1 2 7

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Exercise 3.1

1 Write the modes and median of each set of measures.

a 4 cm, 4 cm, 5 cm, 5 cm, 6 cm, 7 cm

b 51 mm, 4

c 1.2 m, 1.8

d 101 cm, 1

2 What is the

a 5, 6, 7

b 9, 9, 2, 8

c 10, 12

d 2, 3, 4, 5,

3 a Which of these bowlers has the highest mean average bowling score after six games?

	1
Player A	95
Player B	109
Player C	138
Player D	98

b Players B and D both ha
Which do you think is th
convince your partner th
the table in your senten

4 Find the range of each set o

a 1, 2, 3, 4

c 34, 33, 70, 5, 6, 8

e 11, 10, 15, 13, 11

g 91, 105, 116

Tip

You could use a calculator or computer to add and divide the numbers.

5 Find the range in heights of these two groups of children.

Group 1		Group 2	
127 cm	130 cm	137 cm	131 cm
152 cm	138 cm	129 cm	143 cm
135 cm	138 cm	136 cm	143 cm
141 cm		132 cm	

Which group has the l

Describe how the two
because of their differe

6 Kali, Summer, Benji an

While they were practi
how many skips they

Here are their attempts

	1st try
Kali	6
Summer	3
Benji	0
Kyle	4

a Copy and complete

	Rang
Kali	
Summer	
Benji	
Kyle	

b Who do you think h
Explain your answer using the information in your table.

7 Gabriella and Demi have recorded the temperature in the shade at midday on every day of their six week school holiday.

	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Week 1	25 °C	20 °C	20 °C	19 °C	20 °C	24 °C	26 °C
Week 2	25 °C	25 °C	28 °C				
Week 3	28 °C	27 °C	27 °C				
Week 4	15 °C	19 °C	23 °C				
Week 5	18 °C	19 °C	20 °C				
Week 6	32 °C	32 °C	32 °C				

a Design a table to record the three ty
range for each week of the holiday.
the averages and range.

b Which week had the greatest range

c Use the information in your table to
warmest week of the holiday.

d Gabriella and Demi were taking part
on every Tuesday of the holiday. Wh
temperatures on Tuesdays? What w
(mode, median and mean) temperat

8 The four fictional countries of Fratania, Spanila, Brimland and Gretilli celebrate a dry weather festival during the months of January to May. They are each trying to encourage tourists to visit their own countries. Here are graphs of each country's rainfall last year for the five months of the festival.

Fratania

Spanila

Brimland

Gretilli

Think like a mathematician

Daphne the dog had four litters of puppies. The mean average number of puppies in a litter was 5. Investigate how many puppies could be in each litter. Find different ways that make the mean average 5.

What do you notice about the total number of puppies in each solution where the mean is 5?

Check that it is true for another solution. Explain what you find out.

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WALT: Understand that the mode, median, mean and range are ways to describe and summarise data sets.

1 Write the modes and median of each set of measures.

a 4 cm, 4 cm, 5 cm, 5 cm, 6 cm, 7 cm

b 51 mm, 47 mm, 51 mm, 53 mm, 59 mm, 59 mm

c 1.2 m, 1.8 m, 1.1 m, 2.1 m, 1.2 m, 1.8 m, 1.6 m, 1.4 m

d 101 cm, 106 cm, 95 cm, 105 cm, 102 cm, 102 cm, 97 cm, 101

2 What is the mean average for each set of numbers?

a 5, 6, 7

b 9, 9, 2, 8

c 10, 12

d 2, 3, 4, 5, 6

1 a mode: 4 cm and 5 cm
median: 5 cm

b mode: 51 mm and 59 mm
median: 52 mm

c mode: 1.2 m and 1.8 m
median: 1.5 m

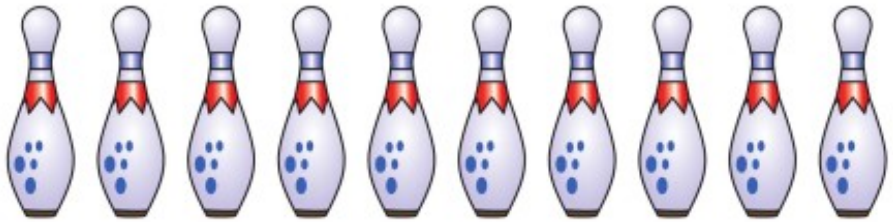
d mode: 101 cm and 102 cm
median: 101.5 cm

2 a 6 b 7 c 11 d 4

WALT: Understand that the mode, median, mean and range are ways to describe and summarise data sets.

3 a Which of these bowlers has the highest mean average bowling score after six games?

You could use a calculator or computer to add and divide the numbers.



	Game					
	1	2	3	4	5	6
Player A	95	108	99	120	95	101
Player B	109	130	124	111	145	131
Player C	138	130	151	157	153	165
Player D	98	154	160	91	129	118

b Players B and D both have the same average bowling score. Which do you think is the better player? Write a sentence to convince your partner that B or D is better. Use information from the table in your sentence.

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3 a Player C has the highest mean bowling score.

b B is better because their scores are more consistent and all their scores are above 100 or D is better because they scored over 150 in two of their games whereas B never scored over 150.

WALT: Understand that the mode, median, mean and range are ways to describe and summarise data sets.

4 Find the range of each set of numbers.

a 1, 2, 3, 4

b 7, 7, 12, 2

c 34, 33, 70, 5, 6, 8

d 26, 21, 35, 63, 30

e 11, 10, 15, 13, 11

f 25, 34, 28, 29

g 91, 105, 116

a 3 b 10 c 65 d 42

e 5 f 9 g 25

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5 Find the range in heights of these two groups of children.

Group 1		Group 2	
127 cm	130 cm	137 cm	131 cm
152 cm	138 cm	129 cm	143 cm
135 cm	138 cm	136 cm	143 cm
141 cm		132 cm	

Which group has the largest range?

Describe how the two groups would look different because of their different ranges.

The range of Group 1 is 25 cm.

The range of Group 2 is 14 cm.

Group 1 has the largest range.

The children in Group 2 are closer in height to each other. The heights of the children in Group 1 are more spread out and there is a bigger difference between the tallest and smallest child in Group 1.

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6 Kali, Summer, Benji and Kyle are learning to skip.

While they were practising they recorded how many skips they did in a row.

Here are their attempts:



	1st try	2nd try	3rd try	4th try	5th try	6th try	7th try
Kali	6	5	6	6	8	11	7
Summer	3	0	3	8	0	7	0
Benji	0	0	1	0	4	0	2
Kyle	4	7	7	6	2	5	4

a Copy and complete this table:

	Range	Mode	Median	Mean
Kali				
Summer				
Benji				
Kyle				

b Who do you think has been most successful at skipping? Explain your answer using the information in your table.

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7 a

	Range	Mode	Median	Mean
Kali	6	6	6	7
Summer	8	0	3	3
Benji	4	0	0	1
Kyle	5	4 and 7	5	5

b Kali has been most successful at skipping because Kali has the highest median and mean averages.

c Learners should indicate the mean or the median. The mode could represent two tries that were much lower or higher than all the others; the mean and the median represent the middle of the data.

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7 Gabriella and Demi have recorded the midday on every day of their six week

	Sunday	Monday	Tuesda
Week 1	25 °C	20 °C	20 °C
Week 2	25 °C	25 °C	28 °C
Week 3	28 °C	27 °C	27 °C
Week 4	15 °C	19 °C	23 °C
Week 5	18 °C	19 °C	20 °C
Week 6	32 °C	32 °C	32 °C

a Design a table to record the three t range for each week of the holiday. the averages and range.

b Which week had the greatest rang

c Use the information in your table to warmest week of the holiday.

d Gabriella and Demi were taking pa on every Tuesday of the holiday. W temperatures on Tuesdays? What (mode, median and mean) tempera

8 a

	Range	Mean	Mode	Median
Week 1	7 °C	22 °C	20 °C	20 °C
Week 2	6 °C	27 °C	25 °C	27 °C
Week 3	8 °C	25 °C	27 °C	27 °C
Week 4	8 °C	18 °C	19 °C	19 °C
Week 5	10 °C	23 °C	28 °C	23 °C
Week 6	8 °C	30 °C	26 °C	30 °C

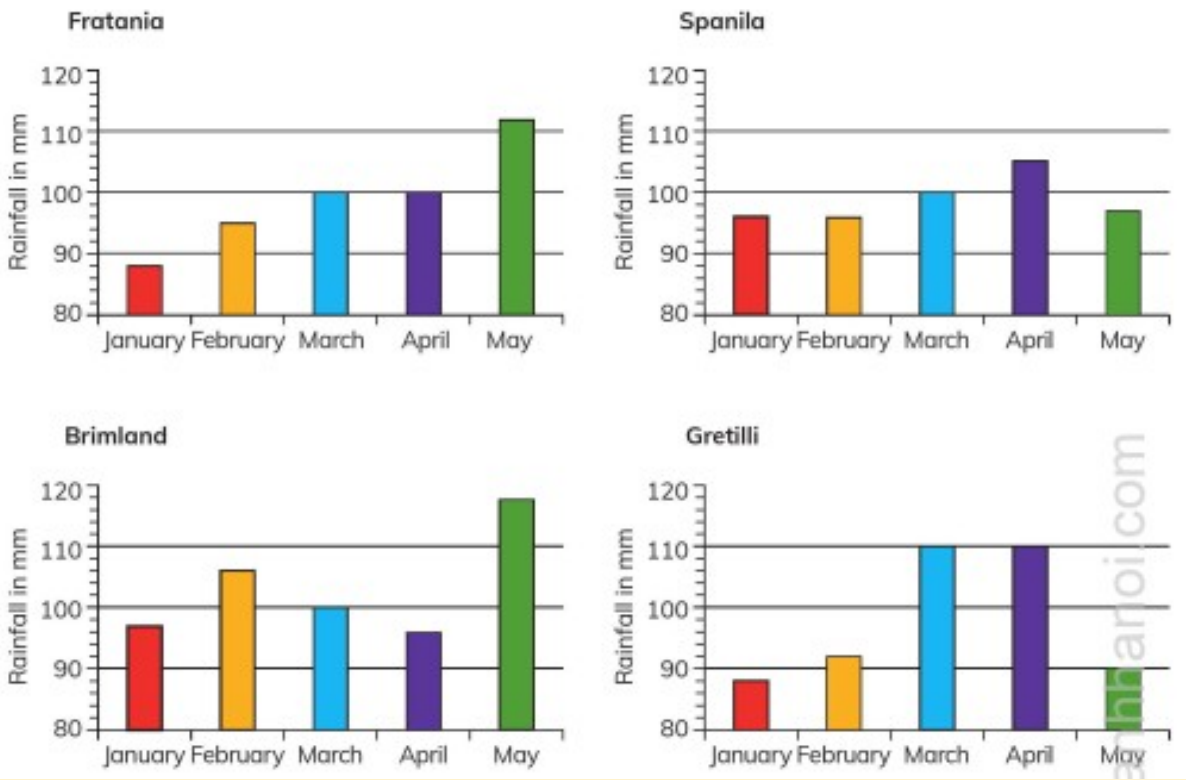
b Week 5

c Learners should use the information from the table to argue which is the warmest, for example: Week 6 is the warmest because it has the highest mean and median, and the mode in Week 6 is the third highest, which also makes it warmer than four other weeks.

d range: 12 °C
mode: 20 °C
median: 25 °C
mean: 25 °C

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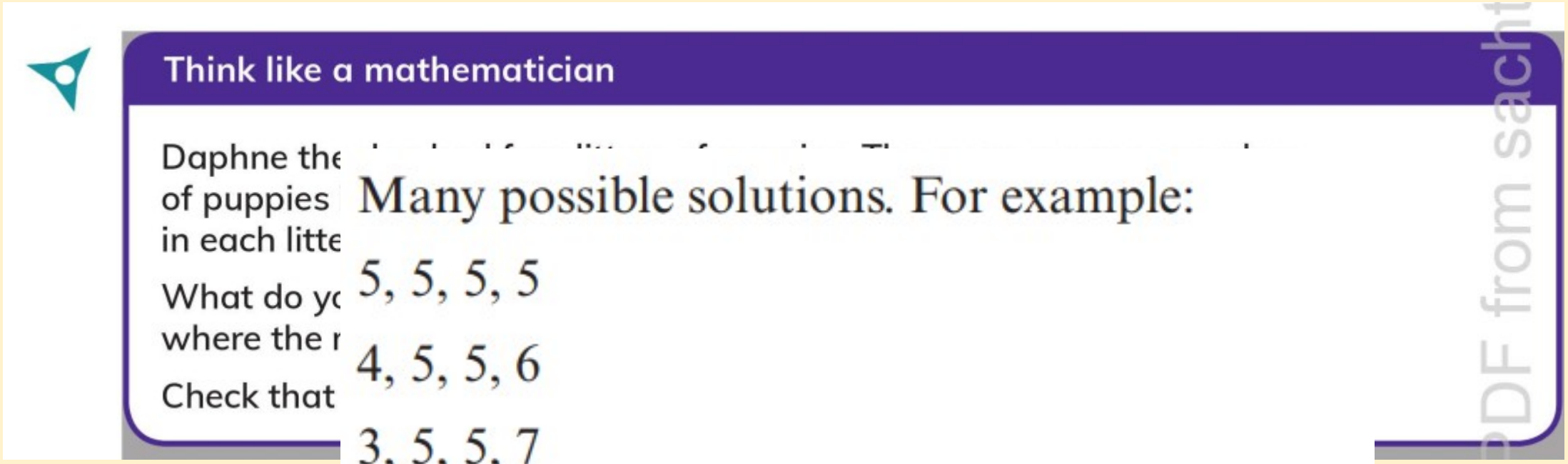
Fratania: mean (99 mm)

Spanila: mode (95 mm)

Brimland: mode (97 mm)

Gretilli: median (92 mm)

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Think like a mathematician

Daphne the
of puppies
in each litte

What do yo
where the r

Check that

Many possible solutions. For example:

5, 5, 5, 5

4, 5, 5, 6

3, 5, 5, 7

4, 4, 6, 6

1, 1, 9, 9

The total number of puppies in each solution is 20, because 20 divided by 4 equals 5.

PDF from sacht

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